

Name: _____ Date: _____

Period: _____

1. The table below shows equally-spaced inputs and their outputs. Complete the **first differences** column.

x	y	Δy (1st diff.)
0	3	—
1	7	_____
2	11	_____
3	15	_____
4	19	_____

The first differences are _____. This means the rate of change is _____, so the data is best modeled by a _____ function.

2. A second data set has first differences of 2, 4, 6, 8. Are these first differences constant? _____

Compute the **second differences** below:

1st diff.	$\Delta^2 y$ (2nd diff.)
2	—
4	_____
6	_____
8	_____

The second differences are _____. This means the data is best modeled by a _____ function.

3. A function's outputs change by roughly the same amount for each unit increase in input. What type of function is the best model? Circle one.

A. Linear **B.** Quadratic **C.** Cubic **D.** Piecewise